Final Practicum Report: *Creator-Specific Comment Intelligence for Optimize Social*

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1. Project Summary

Optimize Social works with creators who receive large amounts of comments across social media posts, but these comments are difficult to translate into clear strategic actions. Although current social media analytics tools (e.g., Instagram Insights and similar reporting dashboards) summarize likes, views, audience reach (views), and overall engagement (number of comments), they do not explain **why** audiences engage or what comments reveal about audience questions, confusion, purchase intent, or content opportunities. As a result, it is difficult for a creator or agency to quickly determine what the audience is discussing, and which themes are most actionable.

This project addresses that gap by transforming raw Instagram comment data into structured, creator-specific audience insights.

To support that goal, our team developed comment intelligence analyses that parse raw comment blocks into comment-level records, clean comment text, create first-pass rule-based signals, aggregate those signals to the post and creator levels, and explore how content characteristics such as topic, timing, and media type relate to audience outcomes. We also incorporated creator-relative performance and viral-post sensitivity checks so that the analysis would not rely only on raw comment counts.

The main purpose of this project is to convert unstructured comment data into structured creator-level insights. Rather than focusing only on broad engagement reporting, the project aims to help answer what followers want, what they are reacting to, and how future content may be improved. The intended value is not just descriptive reporting, but actionable for audience understanding.

The project itself focuses on questions such as:

- What types of content are followers asking for more of?
- What is the emotional temperature of the creator’s audience?
- Which comments suggest confusion or unmet needs?
- What comments indicate product interest or purchase intent?
- What types of comment themes appear most often for a given creator?
- What topics are being discussed from the caption and comments?

- How do topic, timing, media type, and caption style relate to outcomes such as comment volume, question rate, confusion, or purchase intent?
- How are the patterns different across creators?

Overall, the project shows that comment data could be translated into creator-facing outputs such as FAQs, areas of confusion, recurring requests, purchase-intent signals, and opportunity summaries that could support stronger future content decisions.

2. Literature Review

Prior research by Ao et al. (2023) on influencer marketing found that influencer-related characteristics are associated with both customer engagement and purchase intention, which supports treating engagement and purchase intent as related but distinct outcomes. In particular, the Ao et al. report that entertainment value has the strongest association with customer engagement, while credibility has the strongest association with purchase intention, reinforcing the value of analyzing multiple creator-relevant outcomes rather than relying only on one engagement measure (Ao et al.). We used this research as context for framing the analysis, but we did not directly test the causal claims from the prior literature. Instead, our analysis focused on descriptive and diagnostic patterns in the available creator comment dataset.

This broader framing is also consistent with industry guidance on social media analytics, which emphasizes that analytics should not only describe performance but also connect social activity to business outcomes and support better decisions through diagnostic, predictive, and prescriptive uses of data (Newberry).

3. Data Description

The dataset used for this project consists of Instagram-related post and comment data shared through the sponsor’s workflow. The working analysis scope includes 473 posts, 34,881 parsed comments, and 66 creator accounts. The available post-level fields include creator name, caption text, post timestamp, media type, view count, like count, post ID, parsed comments, comment count, assigned niche and topic.

One important characteristic of the data is that comments were initially stored as raw text blocks associated with posts rather than as one row per comment. This meant that a substantial amount of data preparation was required before the comment text could be analyzed systematically. The transformation from raw comment blocks to structured comment-level records was therefore a major early step in the project.

From an analytics perspective, the dataset is useful because it combines textual audience feedback with post-level metadata. This makes it possible to connect general text analysis and comment patterns to content characteristics such as posting time, topic, and media type.

4. Exploratory Data Analysis
    a. Initial Data Characteristics

Initial exploration of the dataset showed substantial variation in activity across creators and posts. Some creators had relatively low comment volume, while others had posts with very high comment counts (Figure 1). This variation suggested early on that raw count-based comparisons across creators could be misleading, and that account-specific interpretation would be important.

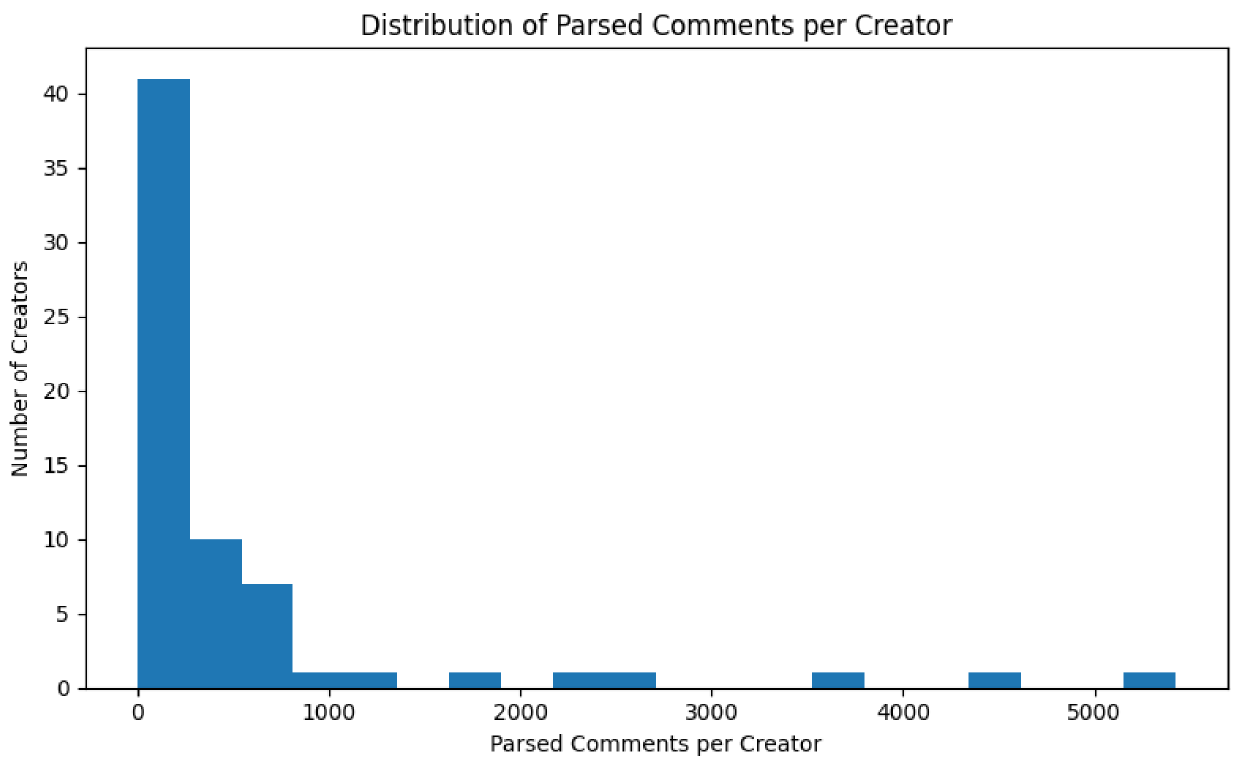

Figure 1. Distribution of Parsed Comments per Creator.

The final working dataset included 473 posts, 34,881 parsed comments, and 66 creator accounts. The distribution of parsed comments per post was highly right skewed: most of those posts received relatively few comments, while a small number of posts generated several hundred or more comments. This pattern suggested that raw comment totals alone could overstate the importance of viral or unusually high-performing posts. As a result, later analysis incorporated the creator-relative performance and viral-post sensitivity checks so that the recommendations would not be based only on raw engagement volume.

Figure 2 shows that comment volume is concentrated among a small number of posts, supporting the need for creator-relative and outlier-aware analysis.

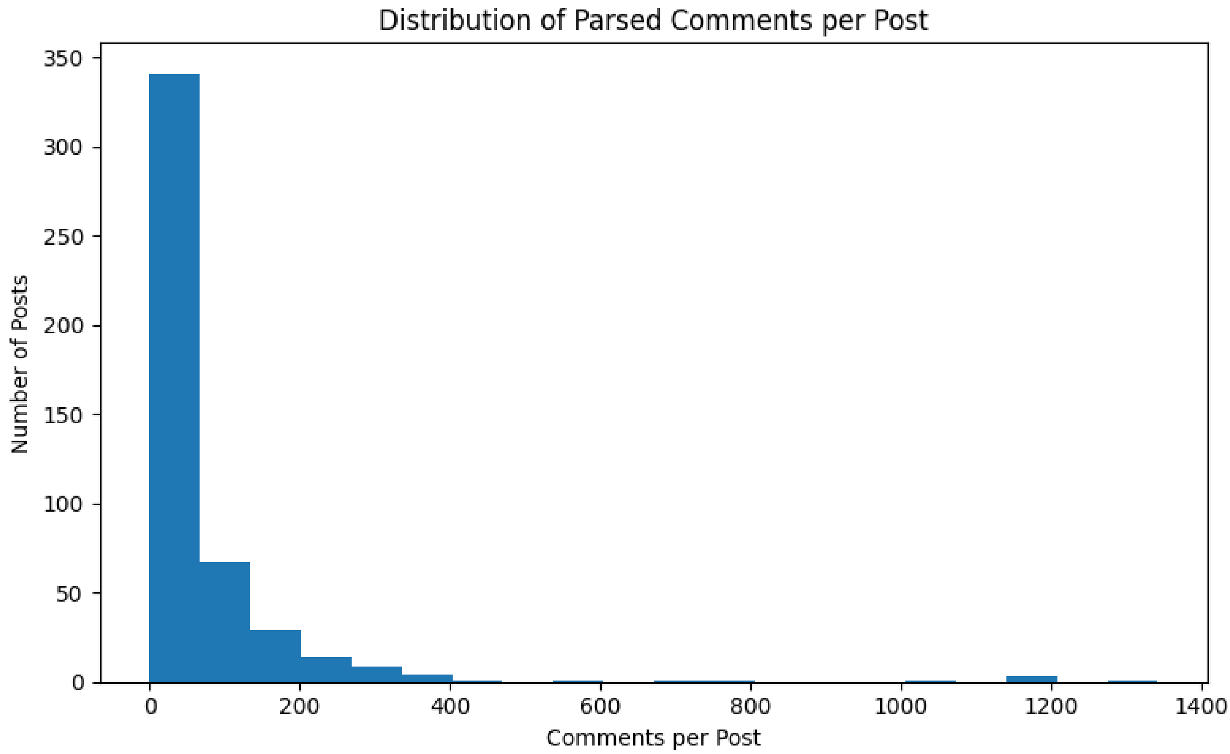

Figure 2. Distribution of Parsed Comments per Post.

b. Signal Distributions

Early rule-based exploration suggested that a meaningful share of comments could be classified into actionable categories such as questions, requests, purchase intent, confusion, and positive reactions. In particular, the frequency of question-like comments suggested strong opportunities for FAQ-style insights and clarification-oriented content recommendations. In later sections we will use other methods, namely LLM-based analyses, to verify these classifications, but the rule-based approach is enough for an initial signal analysis. The exact rules can be found in the appendix.

The initial signal analysis showed that question-like comments were the largest actionable category with 15,142 question comments, representing about 44.7% of parsed comments. Positive reactions accounted for about 11.4% of comments, while product/resource interest, requests, confusion, and pain-point signals appeared less frequently but still provided useful creator-facing insight. This suggests that the strongest immediate use case for the workflow is FAQ generation, clarification-oriented content planning, and product/resource follow-up.

Question-like comments dominate the initial signal categories, which demonstrate that audience questions are one of the clearest opportunities for creator-facing recommendations (Figure 3).

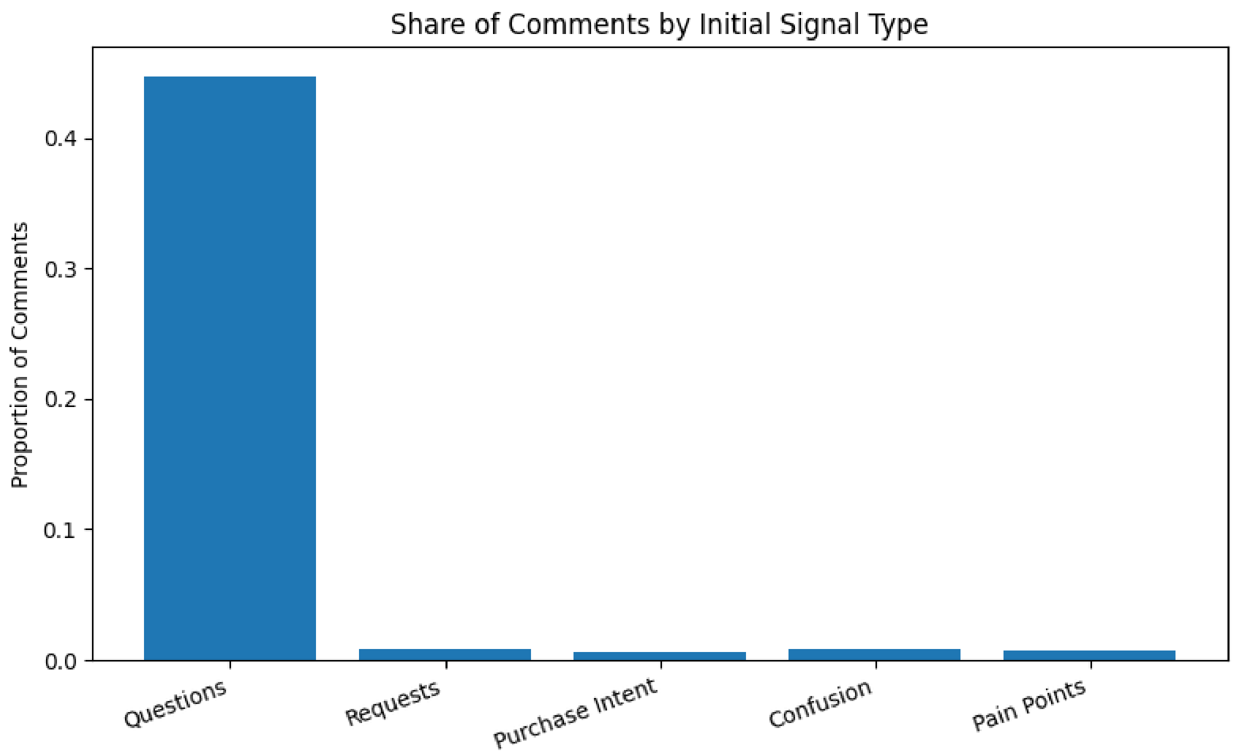

Figure 3. Share of Comments by Initial Signal Type.

Although most comments fall into the broad “Other” category, positive reactions and product/resource interest are large enough to support practical follow-up recommendations (Figure 4).

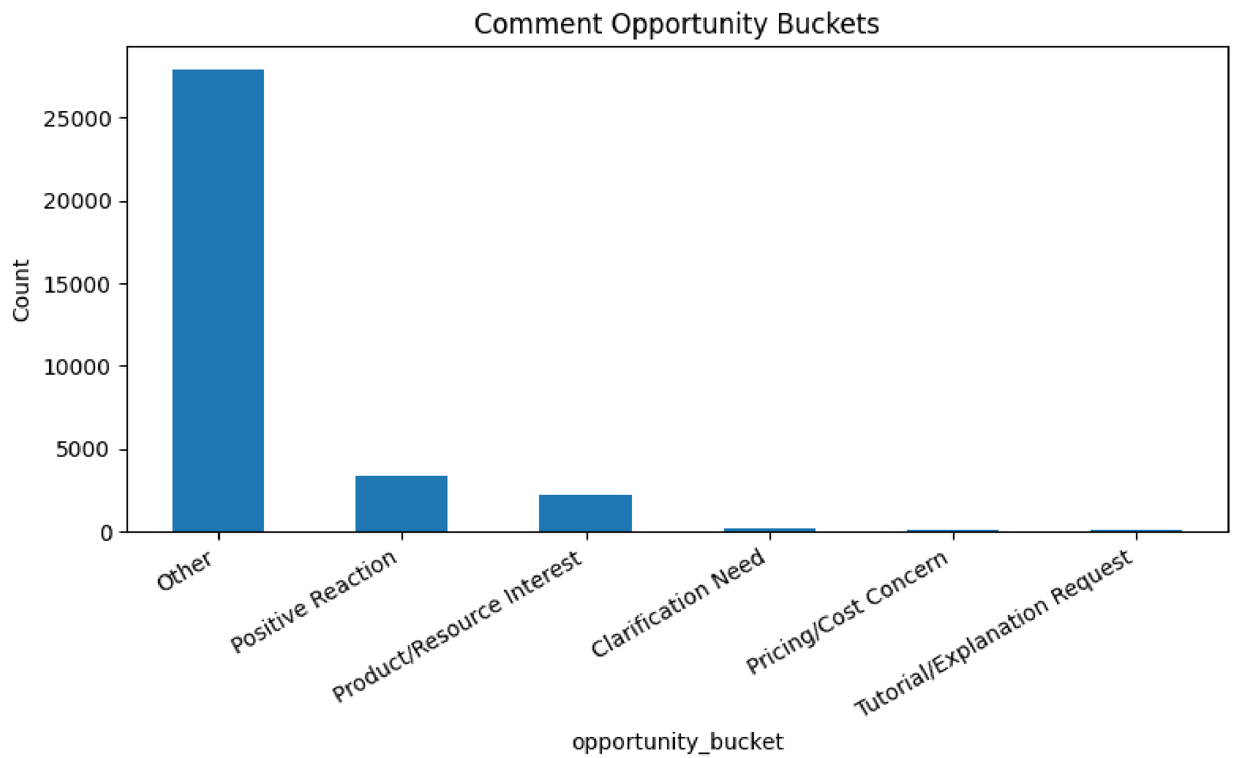

Figure 4. Comment Opportunity Buckets.

c. Variation Across Creators and Posts

The early EDA also suggested that audience comment patterns were not uniform across all creators. Some creators appeared to receive more request-oriented or product-oriented comments, while others received more clarification questions or positive reactions. This reinforced the decision to emphasize the creator-specific outputs instead of relying only on broad average patterns across the whole dataset (Figure 5).

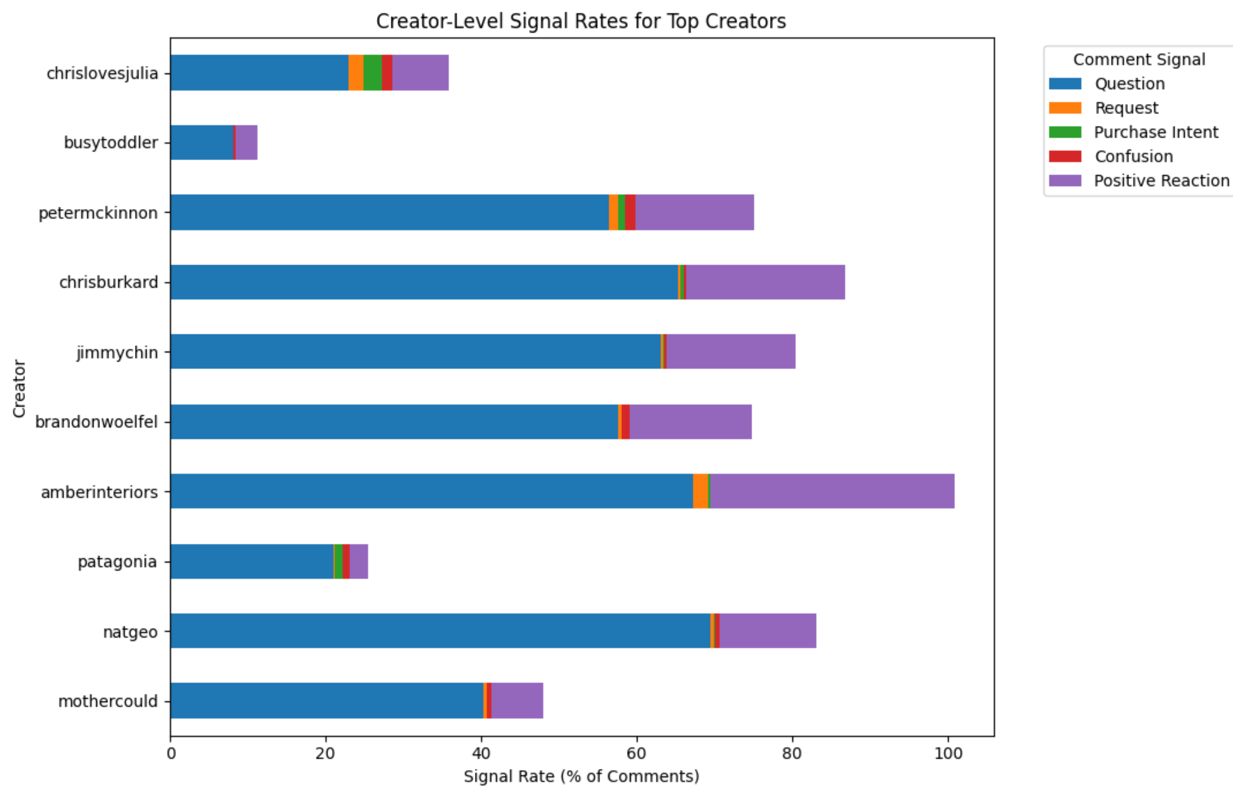

Figure 5. Creator-Level Signal Rates for Top Creators

These creator-level differences show why the aggregate comment patterns are not applied equally to every account. A creator with a higher question rate may benefit more from FAQ-style content, while a creator with stronger product/resource interest or positive-reaction signals may benefit from different follow-up strategies.

d. Early Observations That Motivated the Approach

The EDA phase helped to refine the direction of the project. Rather than treating comments as engagement volumes, the analysis focuses more on extracting the structured audience signals and summarizing them at the creator level. These findings motivated the later emphasis on creator-relative performance, viral-post handling, and creator-facing recommendation outputs.

5. Methodology

After parsing media-level comment blocks into one-comment-per-row records, the comments were then analyzed through feature engineering, sentiment analysis, topic analysis, as well as timing and media-type analysis. These comment-level outputs were aggregated to the post and creator levels, where creator-relative performance metrics and other summary measures were computed to identify meaningful audience patterns and support practical recommendations.

To evaluate the tradeoff between implementation complexity, cost, and classification accuracy, we explored three different approaches: a rule-based regex classifier, a supervised logistic regression classifier trained on sentence embeddings, and an LLM-based classifier. The results were evaluated for alignment with human judgement qualitatively.

We removed low-text, emoji-only, “freebie” comments (viewers commenting a single word in response to the creator’s request to comment that word, in exchange for a benefit or “freebie”), or otherwise unclear comments for the textual analysis to conserve computation time and tokens, but we preserved them as a separate analysis category, in the possibility that these comments may still carry useful audience meaning, even if they are harder to interpret.

a. Comment Parsing and Cleaning

The first major technical step of the project was to parse raw comment blocks into one row per comment while preserving each comment’s connection to its original post and creator. The resulting records were then cleaned to remove usernames, numbering, and formatting artifacts that were not relevant for analysis.

b. Feature Engineering

To identify comments containing questions, requests, purchase intent, confusion, or pain points, we tried three different methods.

We started with a regex-based system to search for certain associated keywords (e.g. “buy” for purchase intent) and determined the comments to contain one or more of the needs based on the presence of these keywords. The exact patterns used can be found in the appendix.

We also trained a logistic regression model on embeddings obtained from Sentence Transformers / SBERT using approximately 100 manually selected examples for each target label (questions, requests, purchase intent, confusion, pain points, as well as 100 control comments containing none of the above). We chose to use Sentence Transformers, all-MiniLM-L6-v2 specifically, because it is fast, offline, and free for our data size.

Lastly, we utilized an LLM model (OpenAI’s GPT 4.1 mini) to classify the comments as containing these needs or not. GPT-4.1 mini was selected because the task primarily involved structured text classification rather than multi-step reasoning, making the smaller model sufficient while reducing cost.

c. Sentiment Analysis

The sentiments expressed in the comments were also explored as part of the broader attempt to understand audience tone and reaction.

To classify whether each comment was positive, negative, or neutral, we used VADER, a rule-based sentiment model that is developed specifically for social media text and designed to handle the short, informal, and abbreviation-heavy nature of online language more effectively than many traditional lexicon approaches (Hutto and Gilbert).

VADER combines a human-validated sentiment lexicon with rules for punctuation, capitalization, degree modifiers, contrastive conjunctions such as "but" and negation, making it especially relevant for social media comments that often include slang, acronyms, emoticons, and other forms of informal expression (Hutto and Gilbert).

To identify more fine-grained sentiment analysis beyond positive or negative, such as curiosity, approval, admiration, confusion, annoyance, or desire, we used the fine-tuned BERT-based model trained on GoEmotions, a large human-annotated dataset that consists of around 58k English reddit comments that are labeled across 27 emotion categories plus Neutral.

d. Topic Analysis

To extract key topics from the comments, we tried two different approaches. First, we looked at all the words in the comments, excluding common stop words using the NLTK Corpus, and ranked them in order of frequency. We also utilized LLM-based topic extraction to identify high-level semantic keywords that could not be captured through keyword frequencies alone.

We also tried two clustering approaches to identify common themes within the clusters. Document vectors were generated using TF-IDF and clustered using K-Means. A second clustering approach used Sentence Transformer embeddings with K-Means to group semantically similar comments.

To identify post-level topics, we also used the LLM to automatically extract topics and emotions from the caption text. Like the feature engineering and sentiment analysis sections, the outputs were evaluated for alignment with human judgment qualitatively.

e. Timing and Media – Type Findings

We also explored how day of week, time of day, and media type relate to comment outcomes. This part of the analysis is important because content performance often depends on combinations of factors rather than topic alone. These findings can potentially support more specific recommendations about when and how creators publish certain content types.

Our focus on timing is also supported by industry benchmarking from Sprout Social, which reports that aggregate engagement patterns vary by day and time and can be meaningfully influenced by posting windows, while also emphasizing that such benchmarks should still be compared against each account's own audience behavior ("Best Times to Post"). That logic aligns closely with this project's use of creator-relative performance, since the broad timing patterns provide useful context, but the creator-specific patterns are more actionable for the actual recommendations ("Best Times to Post").

f. Post-Level Analysis

Once comment-level features were generated, those signals were aggregated to the post level so that each post could be evaluated using summary outcomes such as question rate, request rate, purchase-intent rate, confusion rate, positive-reaction rate, average sentiment, total views, likes, and total comment volume.

This allowed the post-performance to be analyzed as a function of the topic, timing, media type, and caption characteristics.

The project also compared topic patterns with and without viral posts, which helped to test whether some of the results were being driven by a small number of unusually high-performing posts. In cases where conclusions changed meaningfully once outliers were excluded, the creator-relative interpretation became especially important.

g. Creator-Level Analysis

The next steps were to generate creator-level summaries that reflect recurring themes in creator's audience comments. These included FAQ-style summaries, opportunity buckets, creator-specific high-signal comment patterns, and topic summaries that could support practical recommendations.

h. Creator-Relative Performance and Viral-Post Handling

A key design choice was to avoid relying too heavily on raw comment counts. A single viral post could substantially distort a creator's apparent best-performing topic or content type. To address this, the project compared topic performance with and without the viral/outlier posts and used the creator-relative performance measures to evaluate whether a post performed above or below the creator's usual baseline.

i. Key Metrics / Scoring Definitions

The project tracked multiple outcomes separately rather than relying entirely on one blended engagement metric. This was important because a post with lower total comments might still be valuable if it generates high purchase intent, frequent clarification questions, or especially useful audience requests. As a result, the analysis prioritized separate outcomes such as question rate, purchase-intent rate, confusion rate, and creator-relative comment performance.

6. Results, Insights, and Recommendations
    a. Feature Engineering

Table 1 displays the number of comments containing questions, requests, purchase intent, confusion, or pain points out of the remaining 20,615 comments after “freebie comments” and low-text/emoji comments have been removed. Because the methods operate differently, the counts are not expected to match exactly and instead illustrate differences in how each approach interprets the comment text.

Table 1. Summary of Features for the Approaches

	RULE-BASED	CLASSIFIER	LLM
QUESTIONS	9943	7127	3552
REQUESTS	243	7127	3442
PURCHASE INTENT	203	909	838
CONFUSION	253	2711	185
PAIN POINTS	257	3467	110

The three approaches produced substantially different feature counts, particularly for questions and requests. The rule-based approach identified considerably more questions than either machine learning approach. In contrast, the logistic regression and LLM classifiers produced more conservative classifications that better reflected the semantic content of the comments.

Samples of comments classified by each approach can be found in the appendix.

One major limitation of the regex rule-based approach is the inability to understand the context in which keywords appeared. For example, one of the rules for whether a comment contained purchase intent or not is the presence of the word “buy”. Thus, the comment “I will no longer buy from you again!” was marked as having purchase intent when a human would most likely categorize it as the opposite. This is also why the number of questions under this approach is so high, for one of the “keywords” used to determine the presence of a question was the question mark (“?”), but emojis are recorded in the dataset as two question marks (“??”). We were able to filter out comments containing only emojis, but not those using only one.

The rule-based approach also failed to detect the features in comments that didn't fit the pattern. E.g. "Where did you get that chair" was not recognized as having a question, request, or purchase intent due to the limitations of the regex.

The logistic regression classifier trained on embeddings from Sentence Transformer was better at leveraging the semantic representations of the comments compared to the rule-based classifier. However, the manual data curation of 600+ comments (100 comments for each feature, plus 100 "Other" comments that contain none of them) was extremely costly timewise, and the equal target distribution of the training process also likely contributed to the more even distribution of the output compared to those of other approaches.

The LLM was generally better at understanding the context of the comment text without the constraints of the embedding-based classifier. One shortcoming the approach had was, however, the model's lack of understanding of sarcasm, hyperbole, and cultural context. For example, a comment saying: "my hot take is that someone from tropic should adopt me so I don't need to look at all the gray pools around here" is, to most human understanding, not a literal adoption request for an individual residing in the tropics; however, the LLM does not have this human context, and classified it as a Request.

b. Sentiment Analysis

Using the VADER-based sentiment analysis model, our analysis characterized 47.9% of the comments as positive, 11.6% as negative, and 40.5% as neutral (expressing neither positive nor negative sentiment). See Figure 6 as an illustration of the sentiment distribution.

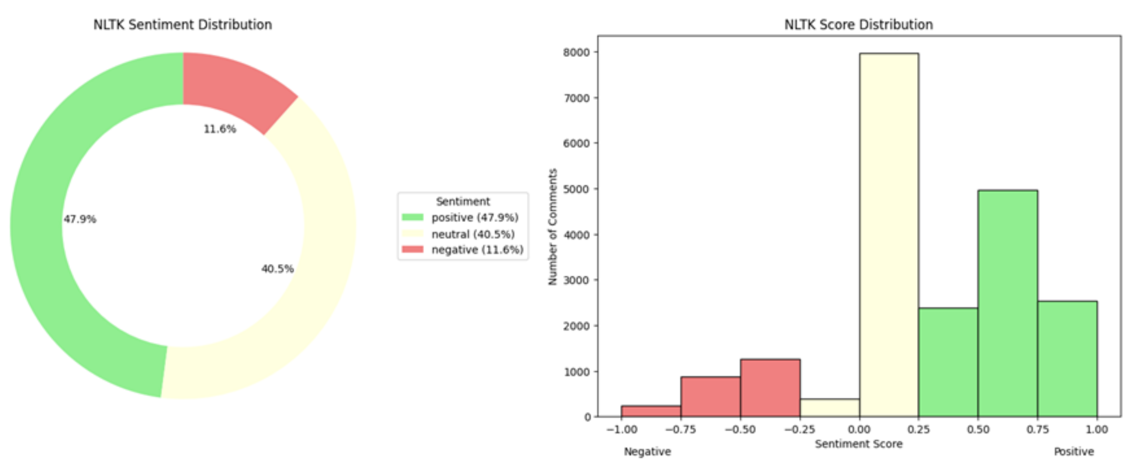

Figure 6. Distribution of Positivity in Comments

This suggests that audience interactions were generally supportive rather than critical. However, sentiment distributions varied considerably across creators. For example, Patagonia received a higher proportion of negative comments compared to the average (Figure 7).

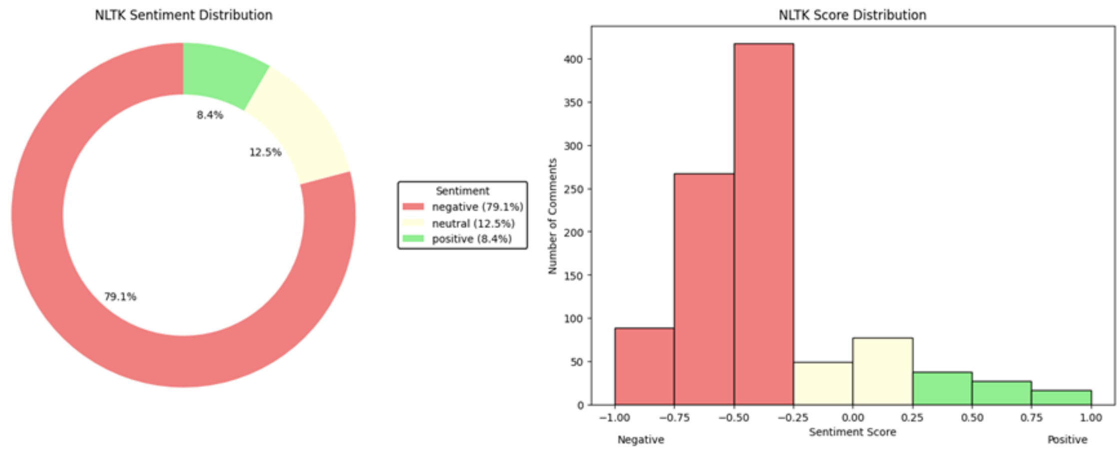

Figure 7. Distribution of Positivity in Patagonia Comments

The finer emotional analysis by GoEmotions revealed that, after removing neutral, admiration was the most frequent emotion expressed in the comments, followed by curiosity, confusion, and love. The least popular emotions were grief, nervousness, and relief. This is in line with the observation in the previous section that most comments are positive or neutral.

The predominance of admiration suggests that many users were expressing appreciation for creators, while the high frequency of curiosity indicates that comments often express interest and request additional information about featured products, locations, or recommendations.

Neutral was removed when charting the emotions because GoEmotions's emotional assignment is not exclusive (i.e. a comment can be assigned to multiple emotions if the text contains evidence for them all). As a result, many comments received both Neutral and one or more emotional labels, making the Neutral category disproportionately common and less informative for comparing the remaining emotions (Figure 8).

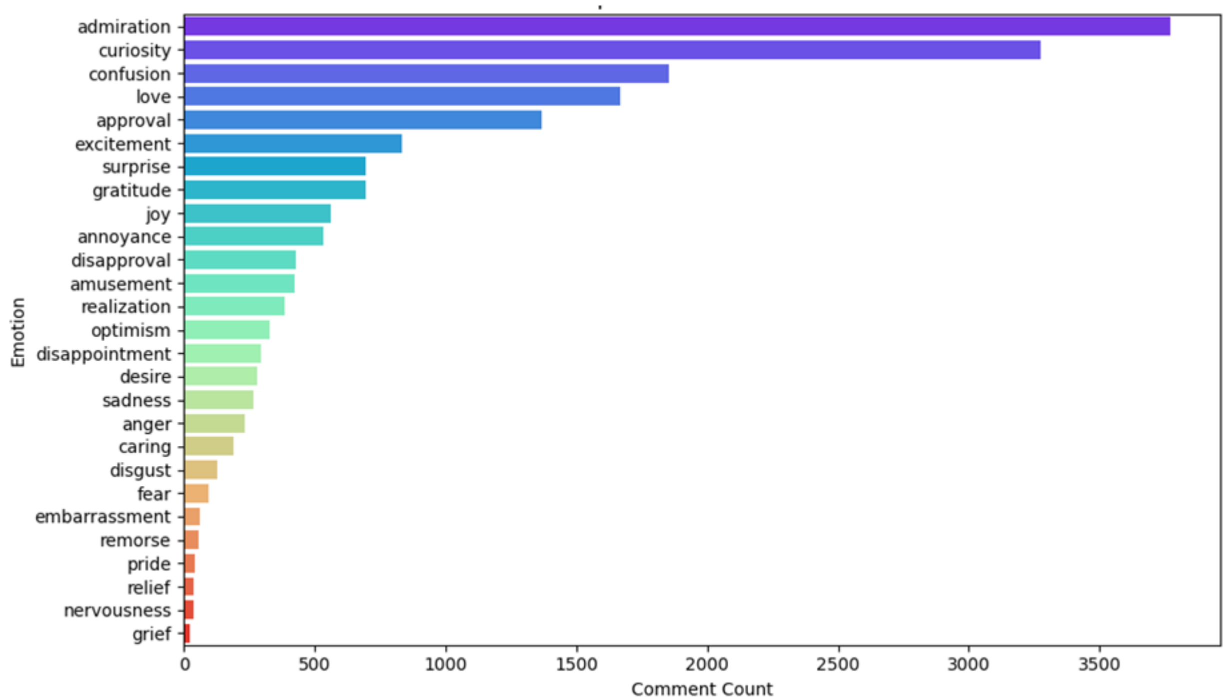

Figure 8: Top Emotion Distributions by GoEmotions (Neutral Removed)

Because GoEmotions performs multi-label classification, the percentages shown in Figure 8 do not sum to 100%.

c. Topic Analysis

See Table 2 for a comparison of the top keywords and topics extracted using a frequency-based keyword ranking (after removing common stopwords with the NLTK corpus) and an LLM-based topic extraction approach.

Table 2. Most common keyword/topic based on frequency ranking and LLM

Keyword (Frequency)	Comment Count (Frequency)	Topic (LLM-based)	Comment Count (LLM-based)
love	1633	Lawsuit	827
drop	969	Person	533
lawsuit	967	Comment	438
like	882	Birthday	344
one	681	Appearance	343
beautiful	643	Link	312
amazing	621	Work	309
great	492	Photo	300
happy	466	Content	298

wow	465	Photos	280

For the frequency-based approach, even though we had tried to filter out common words, many of the highest-frequency terms are generic expressions (e.g., *beautiful*, *wow*) that convey sentiment but provide limited information about the specific topics being discussed. Also, individual words within multi-word phrases are split into separate words (e.g. “drop the lawsuit” becomes “drop” and “lawsuit”) creating redundancy and obscuring meaning.

In contrast, the LLM-based approach produced more semantically meaningful topics than the frequency-based method by grouping related phrases into higher-level concepts rather than treating each word independently. However, there still exist opportunities for further prompt optimization and refinement. First, there are multiple duplicates of the same word in different forms (“Photo” vs “Photos”). Secondly, many of the keywords could be more specific (e.g. “Person”—what person? “Content”—what content?)

To identify recurring themes, K-Means clustering was also applied using both TF-IDF vector representations of the comment text and Sentence Transformer embeddings. A sample of the resulting clusters for one creator (chrislovesjulia) is included in Tables 3a and 3b as a representative sample.

Table 3a. Sample of TD-IDF clusters for chrislovesjulia

Cluster	Terms	Interpretation	Example Comment
2	door,color,paint,trim,recommend,frame,source,leighasecuedesign,oooh,original	Construction Details	What is the paint color of the trim around your front door?
			I would love to know the pocket door source if that detail

			isnâ€™t original ???? Needing this for our laundry room!
			Iâ€™m obsessed with this!! I kind of want to try it on my front door
8	pool,color,blue,dark,cover,live,love,info,float,turquoise	Comments about pools	can I have the link to the pool float?
			My moms pool (Vancouver, BC) is black bottom and it fits in with the cedars and all the green perfectly
			I totally agree. We live in PA and Iâ€™d love a

			darker *still visible* pool vibe. ????????
9	like,run,walk,gorgeous,don,running,instead,legs,absolutely,choose	Comment about animals	As someone whoâ€™s studied birds, birds will walk to conserve energy because flying is energetically costly, especially species that have to jump to start flying like this one.
			like if I had wings..
			But like they also have legs.

Table 3b. Sample of KMeans Embedding clusters for chrislovesjulia

Cluster	Interpretation	Example Comment

13	Comments about pools	white water is clear in the shallow end and fades into a crystal clear blue similar to a white sand beach
		what is the pool color?
		we live in ms and i want a light colored pool so i can see if there are snakes in it ??
8	Asking for information or links	do you have a link for the float? it's adorable!
		inflatable link?
		link for that float? please ??
10	Comment s about projectors	they look horrible upclose when the light hits them terrible ad
		?? ooh! do the links come with projector suggestions?'
		what projector do you use?

Both forms of clustering were able to organize the comments into distinct thematic groups. The embedding-based clusters generally exhibited slightly stronger semantic coherence than the TF-IDF clusters, particularly for information-seeking comments. However, the resulting clusters generally required manual interpretation and were less immediately actionable than the LLM-generated topic summaries, which produced clearer high-level themes with less post-processing.

See Table 4 for the caption topics/keywords corresponding to the highest total views, likes, and comments, as extracted by the LLM, arranged in descending order.

Table 4. Caption topics/keywords corresponding with highest engagement.

Highest Views	Highest Likes	Highest Number of Comments
Window film	POV	Photography
5/19	Overstimulated	Guizhou
Renter-friendly	Long way	TheContainerStore
Launch	Parenting	Natgeo
2 patterns	Mother’s day	Canon USA
Under $30	Self-doubt	Parenting
Miami casting	Mom hacks	Contribution
Photobooth	Mom life	2016 vibes
Smooth	Kids love	2026
Chaos2themax2.0	contribution	thenorthface

The caption analysis suggests that certain content themes—including home improvement (e.g., *window film*), parenting, and photography—were associated with higher engagement.

Other extracted keywords (e.g., *5/19* and *2026*) appear to reflect dates or isolated references rather than meaningful content themes, indicating opportunities for additional prompt refinement or post-processing. However, it is important to note that certain niches and creators receive more engagement in this limited dataset, as discussed further in section 5f.

Future iterations of the system could incorporate additional engagement indicators, such as purchase-intent rates or positive sentiment, to better distinguish highly engaging topics from topics that simply generate large volumes of discussion.

d. Post-Level Signal Analysis

This analysis connects each post topic to the type of audience response it generated, so recommendations are based on what followers are asking for or reacting to rather than comment volume alone. By comparing question rate, purchase-intent rate, positive reactions, and outlier sensitivity across topics, the analysis shows which topics are most useful for FAQs, product/resource follow-up, clarification, or future content planning.

Table 5. Post-Level Topic Signal Patterns

Topic	Main Signal Pattern	Interpretation
DIY/Tutorial	Highest question rate, about 51.7%; highest positive reaction rate, about 18.7%	Strong opportunity for FAQ content, tutorials, and follow-up explanations
Organization	Highest purchase-intent rate, about 1.3%	Strong opportunity for product links, resource lists, and shopping-related follow-up
Renovation	High raw comment performance but sensitive to viral posts	Recommendations should account for outliers before treating this as a consistently high-performing topic
Outdoor/Home Feature	High raw comment performance but sensitive to viral posts	Topic-level averages should be checked against viral-adjusted results.

This post-level comparison shows that different topics generate different types of audience responses. DIY/Tutorial posts appear especially useful for generating questions and positive reactions, while Organization posts appear more connected to product/resource interest. This supports the use of signal-specific metrics rather than ranking posts only by total comment volume.

The current analysis suggests that some posts generate concentrated spikes in questions, requests, confusion, or purchase-related signals even when raw engagement alone does not fully explain their value. This supports the use of signal-based summaries as a more creator-relevant interpretation layer than comment counts by itself.

The final analysis found that question-like comments were the most common actionable signal, with 15,142 question comments across the dataset, representing about 44.7% of parsed comments. Positive reactions accounted for about 11.4% of comments, while product/resource interest, requests, confusion, and pain-point signals appeared less frequently but were still strategically useful. This suggests that the highest-value output of the workflow is not simply identifying "high engagement" posts, but identifying the specific type of audience response each post generates.

The opportunity bucket analysis showed that most comments fell into a broad "Other" category, but several creator-relevant categories were still large enough to support action. The dataset included 3,305 positive-reaction comments, 2,208 product/resource interest comments, 233 clarification-need comments, 143 pricing/cost concern comments, and 80 tutorial/explanation request comments. These categories can help creators prioritize follow-up content, product links, clarifying captions, or tutorial-style posts.

e. Creator-Level Recommendation Outputs

These outputs translate the comment signals into practical creator-facing recommendations, such as FAQs, product links, clarification opportunities, and repeatable content themes. The goal is to make the analysis usable for creator strategy by showing not only which posts received engagement, but what kind of action the creator could take based on the audience response.

Table 6. Example Creator-Level Outputs and Use Cases

Creator-Level Output	What it Shows	How a Creator Could Use It
FAQ Summary	Repeated audience questions	Create Q&A posts, pinned comments, caption clarifications, or story follow-ups
Product/Resource Interest	Comments asking for links, products, sources, or recommendations	Add product links, resource lists, affiliate links, or follow-up posts
Confusion Signals	Comments showing uncertainty, misunderstanding, or need for clarification	Clarify captions, explain steps, add context, or create tutorial-style content

Positive Reaction Themes	Content themes that receive favorable audience reactions	Repeat successful formats, themes, or visual styles
Viral-Adjusted Topic Performance	Topics that remain strong after excluding outlier posts	Avoid basing future strategy only on one unusually viral post

These creator-level outputs translate the analysis into practical recommendations. Instead of only reporting which posts had the most comments, the workflow identifies what kind of audience response each creator is receiving and how that response could inform future content strategy. This makes the analysis more useful for creator strategy reviews, client reporting, and content planning.

Topic-level signal rates showed meaningful differences in audience behavior. DIY/Tutorial posts had the highest question rate at about 51.7% and the highest positive reaction rate at about 18.7%, suggesting that tutorial-style content often generates both curiosity and appreciation. Organization posts had the highest purchase-intent rate at about 1.3%, suggesting that product/resource follow-up may be especially useful for that topic. These differences support the value of analyzing comment signals by topic rather than relying only on total comment volume.

Current exploratory analyses suggest that the comment efficiency and interaction patterns can vary across days of the week, and these aggregate patterns may not always align with creator-level behavior. This reinforces the value of combining high-level timing summaries with creator-relative analysis rather than relying only on one global scheduling pattern.

Table 7. Viral/Outlier Sensitivity by Topic

Topic	Average Comments Including Viral Posts	Average Comments Excluding Viral Posts	Interpretation
Renovation	195.7	36.0	Strongly influenced by outliers
Outdoor/Home Feature	150.2	43.0	Strongly influenced by outliers
DIY/Tutorial	More stable	More stable	More reliable topic signal
Organization	More stable	More stable	More reliable topic signal

This sensitivity check confirms that some apparent high-performing topics were driven by a small number of unusually successful posts. For creator recommendations, raw topic averages should therefore be interpreted alongside outlier-adjusted results. This helps to prevent the report from recommending a topic only because one post performed unusually well.

The viral/outlier sensitivity check showed that some topic-level averages were heavily influenced by a small number of high-performing posts. For example, Renovation posts averaged about 195.7 comments when all posts were included, but only 36.0 comments after excluding viral posts. Outdoor/Home Feature posts also dropped from about 150.2 to 43.0 average comments. In contrast, DIY/Tutorial and Organization posts were more stable. This confirms that creator recommendations should account for outliers rather than relying only on raw average comment counts.

Overall, the post- and creator-level analyses show that comment intelligence is most useful when it separates different types of audience responses instead of treating all comments as the same form of engagement. Question rates, product/resource interest, confusion signals, sentiment patterns, and viral-adjusted topic performance each point to different creator actions. Together, these outputs could help creators decide when to create FAQ content, clarify captions, provide product links, repeat successful themes, or discount results that were driven mainly by outlier posts.

f. Day of week/Time of day

When viewed as an overall aggregate, we see that the best time for achieving the highest Comments-to-views ratio is the weekend (Saturday and Sunday) with mid-week the second-best period. Fridays are the least productive days to post (Figure 9).

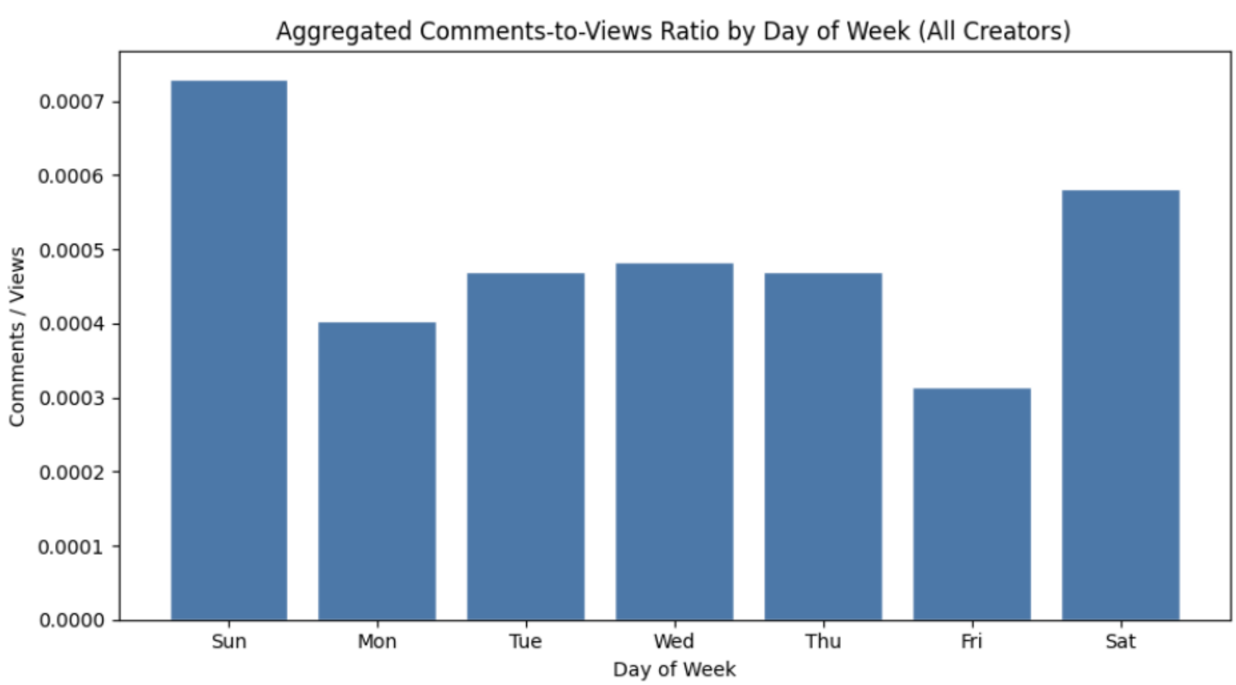

Figure 9. Aggregated Comments-to-Views Ration by Day of Week (All Creators)

However, when segmented by niche, different patterns emerge. The two arbitrarily chosen niches show different overall trends where the audiences comment more on posts made during mid-week rather than the weekend. (Figures 10 and 11) Given the limited number of creator profiles and the small sample size when segmented by niche, no strong conclusions can be drawn. Further analysis is needed on a larger data set.

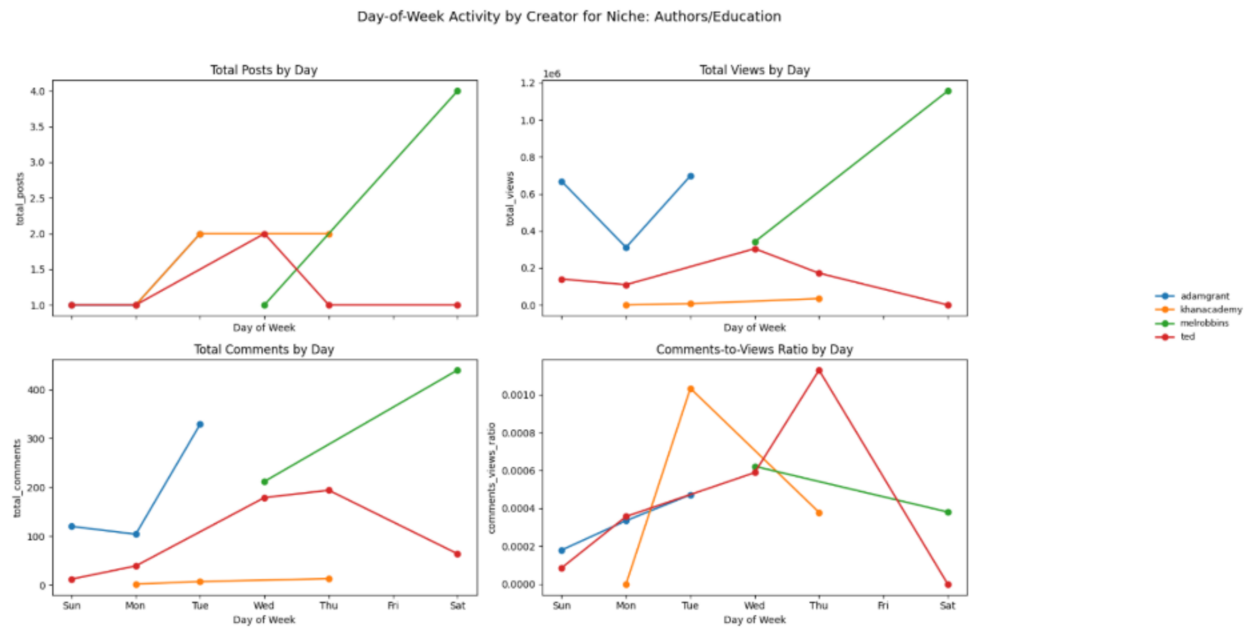

Figure 10. Day-of-Week Activity by Creator for Niche: Authors/Education

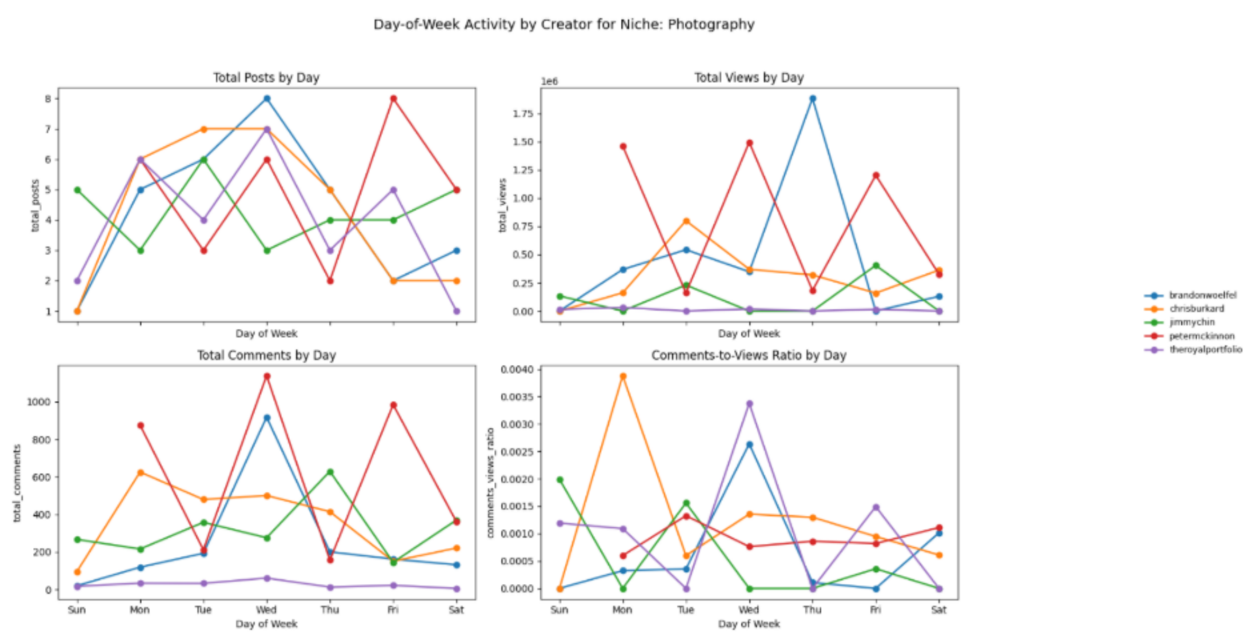

Figure 11. Day-of-Week Activity by Creator for Niche: Photography

Additionally, time of day of post was analyzed to determine best time of day to post for more overall engagement. Whereas the best time of day to post is the 11am hour when viewed in the aggregate (Figure 12), a similar pattern of discrepancy emerges when segmented by niche (Figure 13). As with the day of week analysis, no firm conclusions can be drawn without analyzing a much larger data set.

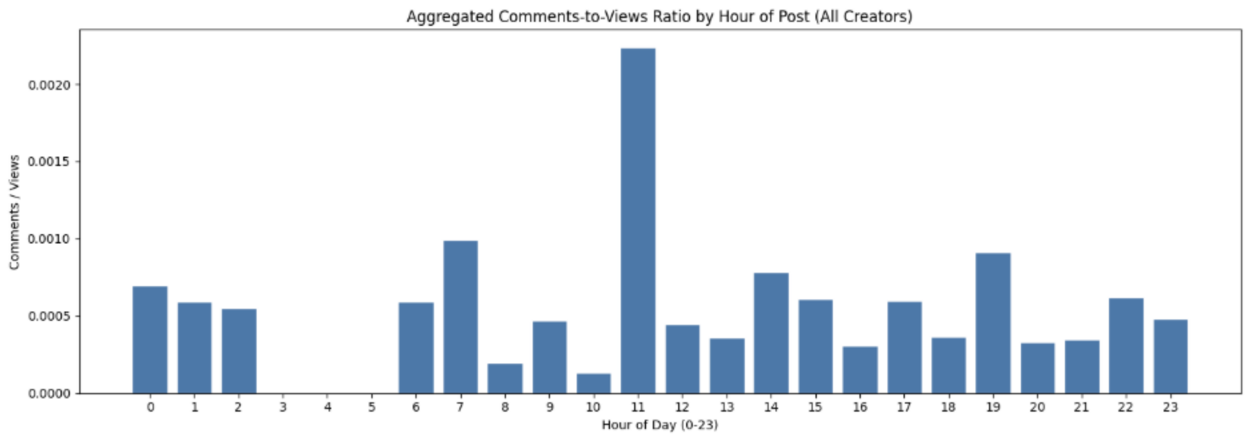

Figure 12. Aggregated Comments-to-Views Ratio by Hour of Post (All Creators)

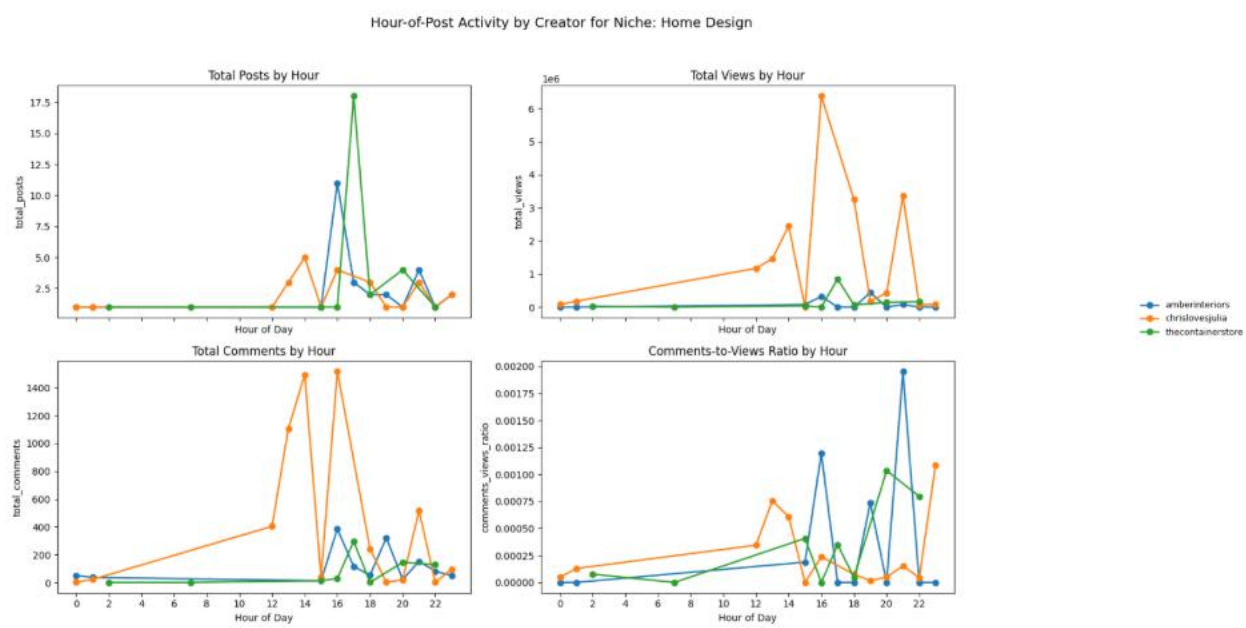

Figure 13. Hour of Post Activity by Creator for Niche: Home Design

Niche is related to topic on a semantic level. Using niche as a categorical variable, various models were examined to predict number of comments and niche was a contributing factor indicating it is relevant to an unknown degree. Therefore, niche should be examined more extensively on a larger data set.

Linear models built to predict number of comments using child_count, day_of_week, time_of_day and media_type (categorical) showed very poor predictive power ($R^2 = 0.0089$) and media_type were not significant. Adding assigned_niche to the model increased the fit by a factor of 10 ($R^2 = 0.0884$), with no significance for media_type and high significance for some of the assigned_niche categories (home design, outdoor/brand, parenting and photography). We conclude that overall media_type is not a significant factor for engagement, but niche can be significant consideration when making content and posting recommendations to clients and should be investigated more on a larger data set.

g. Deployment / Practical Use

A practical deployment path for this work would be an analytics module or reporting layer within the sponsor's workflow or customer dashboard. Possible outputs include creator summary cards, FAQ summaries, audience concern summaries, purchase-intent or confusion alerts, and recommendation-style outputs that suggest what kinds of themes, formats, or post characteristics appear strongest for a creator.

Such outputs could be used in periodic client reporting, creator strategy reviews, or internal content-planning workflows. In this way, the project has potential value beyond the academic setting.

7. Lifecycle Management

If implemented as a continuing workflow, the analysis would need to be refreshed as new comments arrive. Audience language, creator strategy, and audience behavior may shift over time, so the system would need to monitor changes in recurring themes and update creator baselines accordingly.

It would also be important to revisit the performance of the detection rules, particularly if new comment types or platform conventions emerge. Over time, the project could evolve from static analysis into a regularly updated creator-intelligence workflow.

8. Limitations

This project has several limitations. Rule-based methods are useful because they are interpretable and relatively easy to implement, but they may miss some nuance or context in audience language. Some topic assignments may also be approximate rather than exact.

Also, the number of posts and comments varies substantially across creators, which may affect the stability of creator-specific comparisons. Viral posts remain a source of interpretive difficulty even when flagged. Finally, some low-text or emoji-based comments remain difficult to classify in a nuanced way.

Another limitation that we came across is that the current rule-based framework may miss finer-grained emotional nuance in short-form comments, especially when audience reactions are expressed indirectly, through mixed emotions, or through subtle conversational language that does not map cleanly to explicit keywords.

The number of creators and posts vary greatly across the various assigned niches, which skews the overall analysis to the topics and words more common in those communities. This disparity also makes analysis across time and topics by niche problematic due to the various population sizes.

Also, we only had access to the captions as a descriptor for the media content. If we had access to the actual media themselves, further analyses could be performed on the content and other aspects.

9. Future Work

Future work for this project would involve further refining the prompts for higher relevant and actionable outputs, as well as other LLM models (e.g. other OpenAI models, claude, etc.) and embedding models.

Other steps include strengthening caption-topic analysis, improving sentiment or emotion modeling, and exploring whether LLM-based methods provide better creator-level summarization or FAQ extractions than the current rule-based approach. Additional work could also refine the final output layer so that creator-facing insights are easier to consume in a dashboard or report format.

If there had been more time available, the next step would have been to validate the rule-based, embedding-based, and LLM-based classifications against a larger manually reviewed sample of comments. This would make it easier to compare which method best identifies questions, requests, purchase intent, confusion, and pain points in a way that aligns with human judgment. A stronger validation process would also help determine whether the final workflow should prioritize the interpretability of rule-based signals, the scalability of embedding-based classification, or the contextual understanding of LLM-based classification.

Future work could also expand the creator-level output layer by testing the final summaries with actual creator or agency users. For example, the team could evaluate whether FAQ summaries, product/resource interest summaries, confusion alerts, and viral-adjusted topic recommendations are clear, useful, and actionable for content planning. This would help move the project from exploratory analysis towards a practical reporting workflow that could be used in sponsor-facing dashboards or creator strategy reviews.

If stronger emotion modeling produces a much clearer picture of creator-level themes, those outputs could then be compared against LLM-generated summaries to evaluate which method produces the most useful and interpretable creator-facing insights.

Analyzing a larger data set with a larger number of creators and posts in each of the assigned niches should be conducted to refine posting day and time recommendations. Further by removing high volume niche-specific words from the post corpus, a more refined understanding of topics and unique drivers for engagement may be derived. For example, the word ‘image’ in photography may be the essence of the vast majority of posts but may be a significant subtopic for home design or modeling.

Additional future work could include analyzing a larger and more balanced dataset across creators, niches, and posting periods. This would make the timing, media-type, and niche-level findings more reliable and would allow the team to test whether the patterns observed in this project hold across different creator categories. With more time, the analysis could also compare performance over multiple posting periods to see whether audience questions, purchase-intent signals, and confusion patterns change as creator content strategies evolve.

10. Conclusion

This project demonstrates that raw Instagram comment data could be transformed into structured, creator-specific audience insights rather than treated only as surface-level engagement. By parsing raw comments, engineering interpretable features, aggregating those features at the post and creator levels, and incorporating creator-relative and viral-aware analysis, the project creates a foundation for more actionable content intelligence.

The main contribution of the work is its shift from broad engagement reporting to audience understanding. Instead of only asking which posts generated the most comments, the project helps to explain what followers are asking for, where they are confused, where they show purchase interest, and what themes may support stronger future content strategy.

11. References

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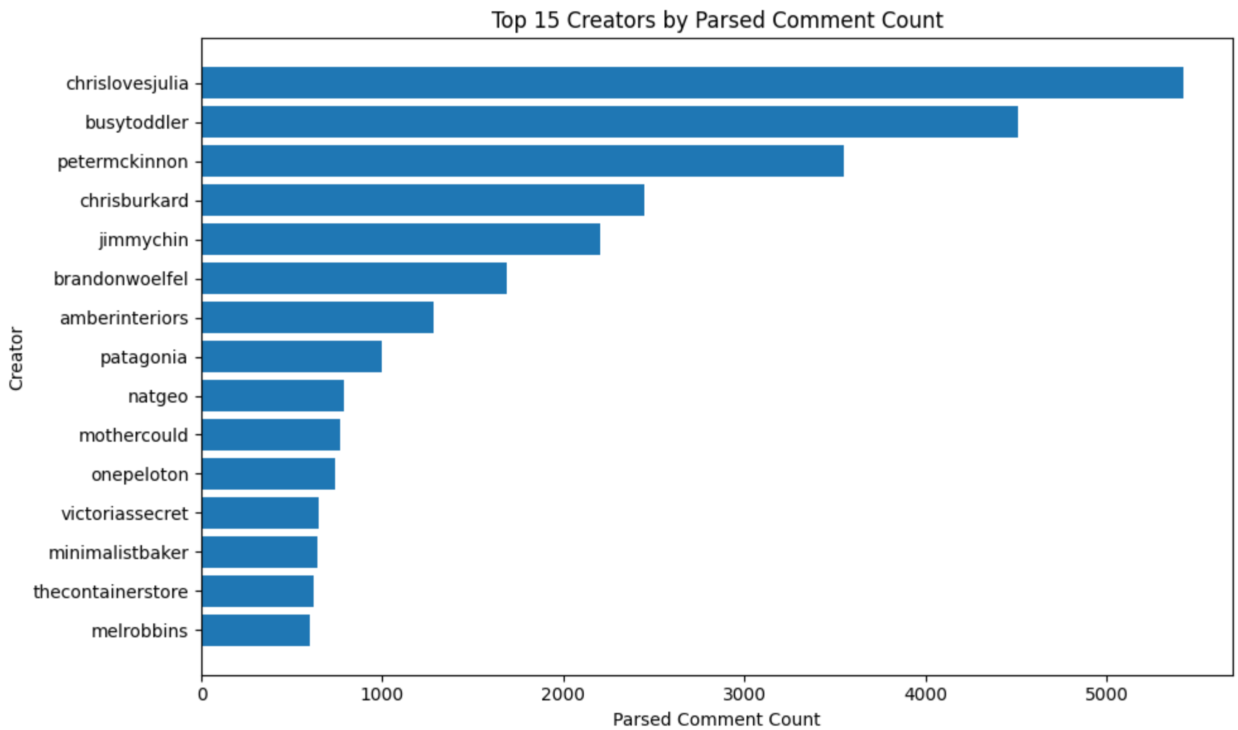
12. Appendix
    a. Extra charts/tables

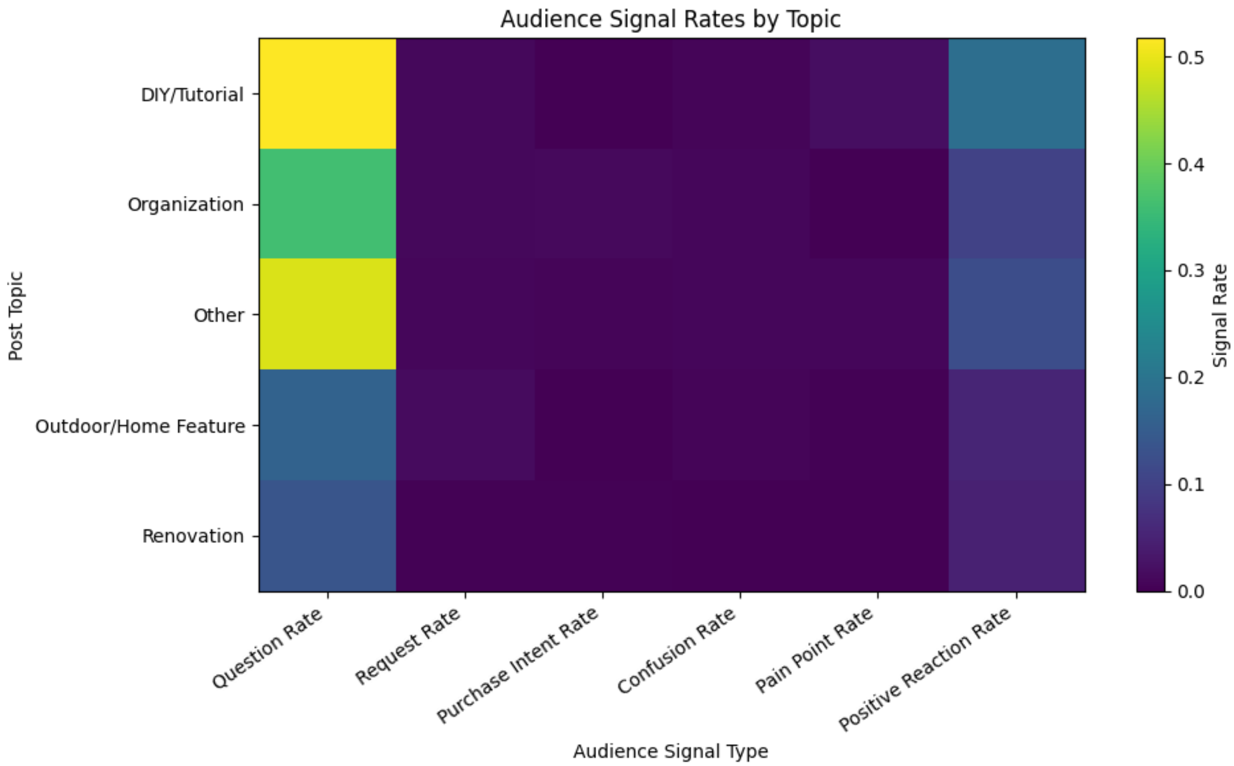
The following charts provide additional exploratory evidence from the notebook that supports the main report findings. The appendix is intentionally limited to a few representative visuals so that the main report stays focused while still showing the supporting analysis behind the conclusions.

Appendix Figure A1 shows the top creators by parsed comment count. This is useful background because it shows that the dataset is not evenly distributed across creator accounts, which supports the report's emphasis on creator-relative interpretation rather than simple raw-count comparisons.



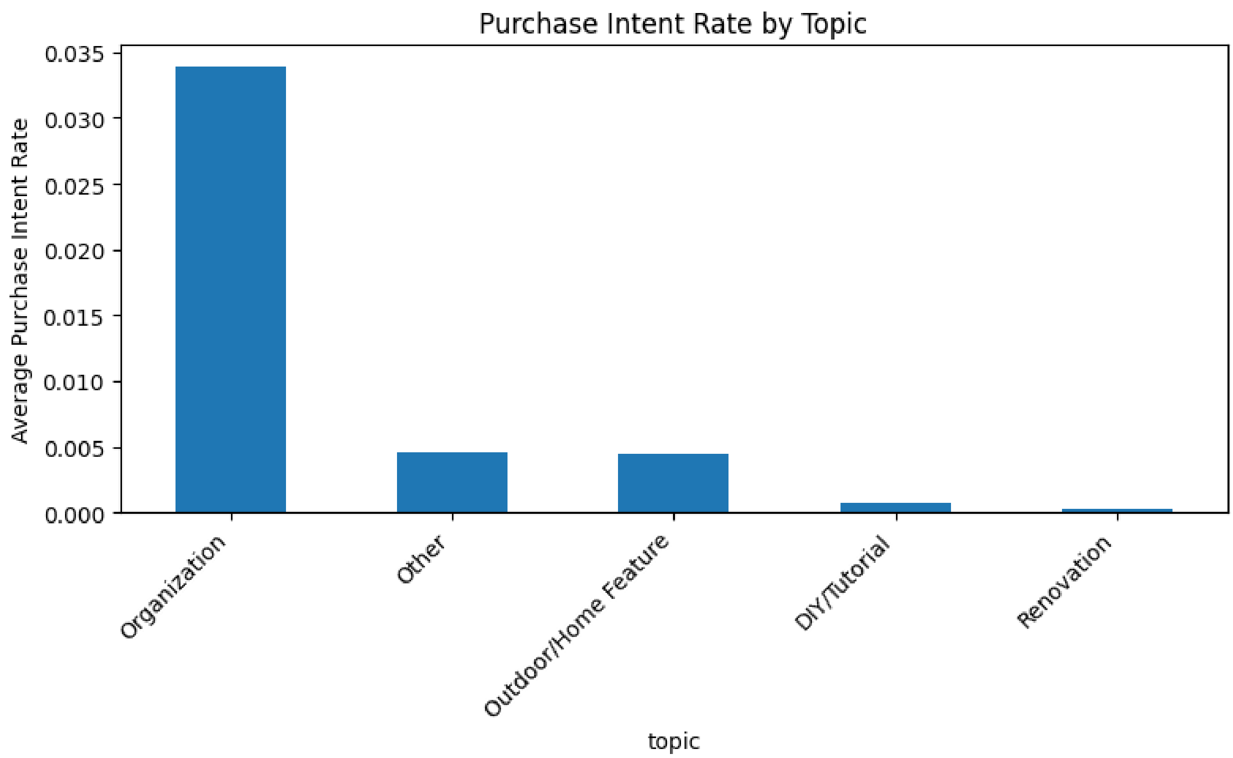
Appendix Figure A1. Top 15 Creators by Parsed Comment Count.

Appendix Figure A2 compares major audience signal rates by topic. This visual supports the post-level signal analysis by showing why topics should be evaluated by the type of audience response they generate, not only by total comments.



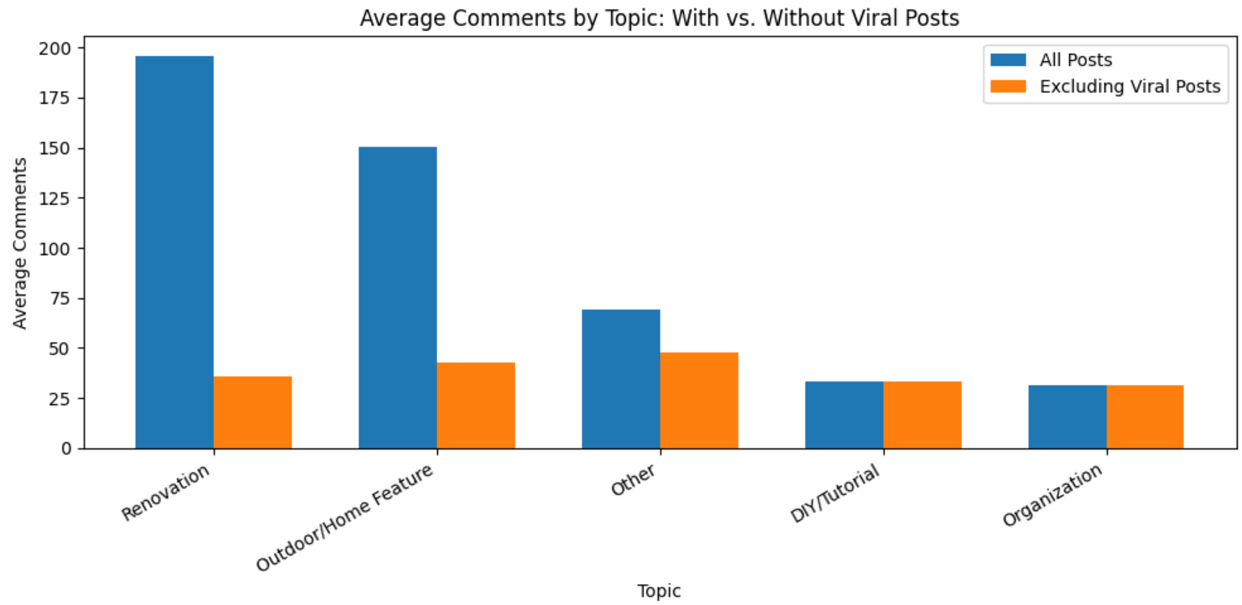
Appendix Figure A2. Audience Signal Rates by Topic.

Appendix Figure A3 provides a focused view of purchase-intent rates by topic. This chart supports the interpretation that Organization posts were more connected to product/resource interest than the other topics.



Appendix Figure A3. Purchase Intent Rate by Topic.

Appendix Figure A4 compares the average comments by topic with all posts included vs after excluding viral posts. This chart supports the viral/outlier sensitivity discussion by showing that some topics looked stronger mainly because of a small number of unusually high-performing posts.



Appendix Figure A4. Average Comments by Topic With vs. Without Viral Posts

b. Example rules

The rule-based labels were created using keyword and pattern matching. These rules were useful for creating interpretable first-pass signals, but they may miss context or incorrectly flag comments when keywords are used sarcastically, negatively, or in unrelated ways.

Appendix Table A1. Example Rule-Based Signal Patterns.

Signal	**Example rule-based keywords/patterns**
Question	?
Request	Can you; please make; want more; would love; show us; tutorial; how to; part 2; next video
Purchase Intent	Buy; purchase; order; get one; where can I get; where do I buy; link please
Confusion	Confused; do not understand; what do you mean; why; not sure
Pain Point	Problem; issue; hard; difficult; struggle; frustrated; annoying; stress; need help
Positive Reaction	Love; amazing; great; awesome; beautiful; helpful; perfect; incredible
Negative Reaction	Hate; bad; awful; terrible; annoying; frustrated; confused

c. Sample creator-level outputs

The sample creator-level outputs below illustrate how comment signals could be converted into practical recommendations for creator strategy reviews or client reporting.

Appendix Table A2. Sample Creator-Level Recommendation Outputs.

Creator Pattern	**Strongest Signal / Pattern**	**Example Recommendation**
Creator with many repeated questions	High question rate	Create FAQ content, pinned answers, or follow-up tutorials.
Creator with product/link requests	Product/resource interest	Add product links, source lists, or shopping-related follow-up.
Creator with unclear audience responses	Confusion signals	Clarify captions, add context, or explain steps more directly.
Creator with strong approval themes	Positive reaction themes	Repeat successful content formats, themes, or visual styles.
Creator with one unusually high-performing post	Viral-sensitive topic performance	Avoid basing strategy only on one unusually high-performing post.

d. Example feature engineer classification

The below tables illustrate the comments categorized into each category according to the different feature engineering approaches:

Appendix Table A3. Sample Question Comments According to Different Approaches

Rule-based	Classification	LLM
`Beautiful!????`	`Savannah has gross muddy ocean water and we are putting in a pool ?? curious where you would land on that spectrum`	`What about brackish brown with muddy water? ??`
`Wait!!! Whaaat?!! Genius!`	`What is the pool color?`	`What vine is that climbing the brick??`
`Set the temp, turn the fountains on, forget the rest ????`	`Need to know where the floaty is from ??`	`Is that pebble tec?`
`My mom taught me this! ??`	`what city is your home in ? nice look`	`Do you have a link for the float? It’s adorable!'`
`do you have a link for the float? it's adorable!`	`I prefer lighter plaster colors so I can see if there's any critters on the bottom`	`How are you keeping your white furniture so crispy white ?? I love the`

		aesthetic but just avoid it with kids

Appendix Table A4. Sample Request Comments According to Different Approaches

Rule-based	Classification	LLM
omgosh a dream kitchen layout and chris knows how to make the most of it! perfect setup love to see it! ??	Link pleaseeee	Do you ship to Australia?
to be in the photo too and have someone who knows how to take a picture take it ??	Amazing!! Buying the diamond pattern ?? How do you recommend addressing if the area I'm covering is wider than 24? Will a seam be noticeable?	Was just looking for these the other day. Link please!
i would love to know the pocket door source if that detail isn't original ???? needing this for our laundry room!	How's Chris feeling?	What is your waterline tile?
love the office of course but can you do a tutorial on how you made the video and with what ??	What cabinets did you use?	Pool liner info ?
yep i knew this also i know how to fold a fitted sheet very neatly too ??	Oh what time is dinner? Hope I didn't miss it??	I'm trying to decide what color for our above-ground pool. Our yard is adjacent to a shopping center and the Best Buy sign glows into our yard at night so I think the blue/yellow logo reflecting on blue would be kind of cool. Any thoughts?

Appendix Table A5. Sample Purchase Intent Comments According to Different Approaches

Rule-based	Classification	LLM
i'll keep this in mind when i go buy my kiddie pool tomorrow	sold me! will be getting this instead of the pocket 4	can we all just do whatever the heck we want?
was just looking for these the other day link please!	oh god im buying this!	take my money
i'm obsessed!! i went to order see the lattice is on back order until	adds to cart for father's day	soooo i may or may not have purchased 35 of these ??????

august i can't wait! totally worth the wait! ??????",		
spotted the trend for similar titles and imagery a few years ago it just feels like a way to trick people into buying the wrong book after one title is getting lots of praise then people buy the other one not realising ??	??????????????????l love the app it's amazing highly recommend!! so worth the money ????	oh my gosh I need the diamond one! ??
why do u talk to this guy and buy his course this is common sense what does he know that u don't	i already bought it this afternoon all 16!!! and i cannot wait we loved playing preschool and i have no doubt we'll love offshoots too!!! ??	I want this!

Appendix Table A6. Sample Confusion Comments According to Different Approaches

Rule-based	Classification	LLM
why complicate life with pool colour?	hmmm i live in southern france not far from the mediterranean sea i'm not allowed to have a turquoise pool? because i do ??	did this collab overtake the kirkland's one?
hahahaha yes why does this make so much sense but i've never realized it ?!	isnt it too cold?	are you raising your prices 30 to accommodate the 20 off coupons?
agree! we lived in dallas tx and hubs really wanted bright and lite it was pretty but not sure what color reflects dallas	omg what!!!! i've been doing it wrong this whole time hahaha	are you no longer carrying office supplies or gift wrap?
our last house the pool looked turquoise because of the reflection of the sky but the pool itself was white that's why so many pools look turquoise	whattt????? 5 years?????no way ???????	ok not seeing the containers of varying sizes for storage purposes
yes!! i been hearing rumors but was not sure it was actually out yet so yea!!	i'm trying to figure out the interior trench	but is it container store pricing or bed bath and beyond pricing they're different

e. Prompt for LLM

The prompt used for the LLM is exactly as below:

```
"""
    There are exactly {len(batch)} Instagram comments.

    For each comment, return:
    - topics: up to 5 specific noun keywords. Exclude emotion words.
    - demands: creator-directed requests/questions, or null if none.
    - category: one of "Request", "Purchase Intent", "Confusion", "Pain Point", or "Other".
    - emotions: up to 5 emotion words, or ["Neutral"] if none.

    Return ONLY this JSON array:

    [
     {{"index":0,"topics":[...],"demands":null,"category":"Other","emotions":["Neutral"]}},
     {{"index":1,"topics":[...],"demands":["..."],"category":"Request","emotions":["curious"]}}
    ]

    COMMENTS:
    {numbered_comments}
    """
```